

Roberto Nogueira

Senior Software Engineer | Ruby on Rails | AI-Native Engineering | Systems Integration

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CONTACT

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PROFESSIONAL PROFILE

Senior Software Engineer with a strong telecom background and **7+ years** of **Ruby on Rails** experience. Enterprise integration, backend architecture and software modernization. International delivery across **OSS/BSS, payroll and billing.** Creator of **ZOAT** and an evolving **AI-native ecosystem** connecting human intent, semantic understanding and AI-assisted realization.

7+ years
of Ruby on Rails

300%
Productivity improvement at Ericsson

International delivery
LATAM • USA • Canada • Europe

PORTFOLIO

enogrob.github.io/projects/

Projects connecting software architecture, systems integration and AI-native engineering.

HOLOFLUX

Knowledge, intent, understanding and creation in AI-native systems.

github.com/enogrob/project-holoflux

MOP AUTOMATION

Ruby automation for telecom. Documented 300% productivity improvement.

github.com/enogrob/project-automation-mop

AI-NATIVE ECOSYSTEM

ZOAT / IOP

ZOAT expresses intent and understanding and implements the IOP paradigm.

COSMOS / HOLOFLUX / FACTORIES

Knowledge organization, semantic understanding, content and automation.

CORE CAPABILITIES

Languages & frameworks

Ruby, Rails, Python, JavaScript, React, Elixir, Phoenix, Erlang/OTP.

Architecture & data

REST, GraphQL, Packwerk, microservices, Kafka, RabbitMQ, PostgreSQL, MySQL, Oracle/PLSQL.

Cloud & delivery

AWS, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD.

Quality & processing

RSpec, Minitest, Redis, Sidekiq, Datadog, Kibana, Rollbar.

AI engineering

GitHub Copilot, OpenAI Codex, Claude Code, context engineering, agentic workflows, multi-agent systems, AI-assisted SDLC, AI-native, ZOAT(IOP).

EXPERIENCE & CREDENTIALS

Software • Integration • Telecom • AI-native engineering



PROFESSIONAL EXPERIENCE

Zoatworks | Jun 2013 - Present

Founder & Software Engineering Consultant | AI-Native Engineering

Independent consultancy alongside full-time roles.

• Developing ZOAT, IOP, HOLOFLUX, COSMOS and FACTORIES for semantic understanding, knowledge organization and human-AI collaboration.

• Event-driven Rails microservice and Kafka infrastructure for Qeevo student subscriptions and API integrations.

• AI-assisted SDLC, context engineering and technical-debt analysis; Ruby, Elixir and Erlang/OTP services; SAP and Oracle PL/SQL integrations.

Projects: AI-native ecosystem; Anubis/CX - Qeevo.

Encora Inc. | Jun 2021 - Sep 2024

Senior Ruby on Rails Engineer | System Specialist

Remote

• Justworks payroll, billing and Employer Risk using Rails, React, GraphQL, Sidekiq and Redis.

• Workers' Compensation billing rules, reconciliation and bulk file workflows with AWS S3.

• Packwerk modular monolith; CI/CD, observability and mentoring; Wind River cloud and edge initiatives.

Projects: PEO / Payroll Platform; Time Tracker - Justworks; WRCP - Wind River.

InMov | Jul 2018 - Jun 2021

Senior Ruby on Rails Engineer | Full-Stack

On-site

• Smart City and public-sector applications using Rails, JavaScript, jQuery and AJAX.

• Led Rails 4.2 to 6.0 upgrades; created Site Rails Manager with Bash/Docker.

Projects: City Hall Permit System; Site Rails Manager.

Ericsson | Dec 2004 - Jun 2013

Software Developer | Integration Engineer | Solution Integrator

On-site

• R&D;, customer support, integration and deployment in large-scale OSS/BSS environments.

• International projects in Canada, Ireland, USA, Mexico and Chile.

• MOP Automation with Bash, Ruby and Rake: 300% productivity improvement; Revenue Assurance for Vivo.

Projects: BSS Dual Stack; Full IP Rehoming; MOP Automation; Revenue Assurance - Vivo.

CREDENTIALS

EDUCATION

Data Communications - USP, 1998

Software Engineering, postgraduate - UNIVAP, 1994

Telecommunications / Electronics Engineering - INATEL, 1985

CERTIFICATIONS

SAP Integration - Thomson Reuters Brasil, Dec 2013

Solution Integrator Experienced (Job Stage 5) - Ericsson, Sep 2008

SPECIALIZATIONS

2026 - Claude Code (Udemy); GitHub Copilot CLI (GitHub)

2025 - AI Automation; MCP Crash Course (Udemy)

2017 - Erlang, University of Kent / FutureLearn

2016-2017 - Rails / Web Development, Johns Hopkins / Coursera

2015 - Web Application Architectures, University of New Mexico / Coursera

2014-2015 - Data Science / R, Johns Hopkins / Coursera

LANGUAGES

Portuguese: Native

English: C2 (Proficiency)

Spanish: Intermediate